

SAVANNA

SENTINEL



Software Requirements Specification (SRS)

Last Revision: 28 July 2026

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Introduction

Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for **Savanna Sentinel**, a wildlife conservation anti-poaching intelligence platform. It serves as the authoritative requirements reference for the development team, the industry client EPI-USE, and university evaluators.

Scope

Savanna Sentinel enables game reserve rangers, analysts, and community liaisons to:

- Detect poaching hotspots through an AI-driven geospatial risk heatmap
- Plan optimal, constraint-based patrol routes
- Capture and sync field reports offline via a Progressive Web App (PWA)
- Submit community tip-offs about observed incidents or sightings
- Administer user accounts and monitor system activity via an audit log

The system is a multi-role web application with offline-first field capabilities, backed by a versioned REST API, a PostgreSQL/PostGIS database, and an asynchronous ML processing pipeline.

Product Perspective

Savanna Sentinel is a standalone web-based platform. It exposes a versioned REST API (`/v1/...`) consumed by a React PWA client. The system is containerised using Docker Compose and deployable to any Linux host via a single `docker compose up` command. An asynchronous Celery worker pipeline handles ML workloads independently of the API request cycle. All external traffic passes through a Caddy reverse proxy for automatic TLS termination.

Assumptions

- Admin accounts are provisioned directly at the database level. No API endpoint creates an Admin.
- The Resend SMTP relay (free tier, 100 emails/day) is sufficient for password reset email volume in the demo environment.
- A computed heatmap must exist before patrol route generation can proceed

Definitions and Acronyms

Term	Definition
RBAC	Role-Based Access Control - access governed by the user's assigned role
JWT	JSON Web Token - stateless, signed authentication token
PWA	Progressive Web App - web application with offline and installable capabilities
PostGIS	PostgreSQL extension for geospatial data types and spatial indexing
SHAP	SHapley Additive exPlanations - ML model explainability framework
MinIO	S3-compatible open-source object storage
IndexedDB	Browser-native structured storage API used for offline data
Workbox	Google library for service worker and offline caching management
Field Report	A ranger-submitted record of an observed incident or sighting

Tip-off	A community liaison—submitted record of a suspected incident or sighting
Geospatial Event	A database entity with a geographic location and timestamp (Incident, Sighting, PatrolTrack)
Soft-Delete	Marking a record as deleted without physically removing it from the database
JTI	JWT ID - a unique identifier embedded in a token used for revocation tracking
AGSI	Asynchronous Server Gateway Interface - Python async web server standard
TLS	Transport Layer Security - protocol for encrypted network communication

User Stories

Identity & Access Management

ID	User Story	Acceptance Criteria
US1.1	As a prospective user, I want to register for an account and select my intended role so that I can request the appropriate level of access once approved.	Form includes username, email, password, first name, last name, and a role selector.

		• Available roles to select are: Ranger, Analyst, Community Liaison. • Password must be at least 8 characters. • Account is created with <code>is_active: false</code> and cannot be used to log in until an Admin activates it. • Returns <code>201 Created</code> on success.
US1.2	As an Admin, I want to review and activate or deactivate user accounts so that I can control who has access to the system.	• Admin can view a comprehensive list of all system users including those pending activation. • Admin can activate, deactivate, or update the role of any user account (excluding other admins).
US1.3	As a registered user, I want to log in with my credentials so that I receive a secure JWT session.	• System validates email and password. • Issues a signed JWT access token and refresh token on success. • Access token expires after 3600 seconds.

		• System restricts login to users where <code>is_active</code> is true. • Any failed login attempt returns a single vague <code>401</code> response that does not reveal whether the email exists, the password is wrong, or the account is inactive.
US1.4	As an authenticated user, I want to refresh my session so that I am not logged out while actively using the system.	• Client can exchange a valid refresh token for a new access token. • Returns <code>401</code> if the refresh token is invalid or revoked.
US1.5	As a user, I want to update my profile and password so that my account information remains current and secure.	• User can update first name and last name. • Password change requires <code>current_password</code> and <code>new_password</code>. • <code>new_password</code> must be at least 8 characters. • Password change revokes all existing refresh tokens.

US1.6	As an authenticated user, I want to log out of the system so that my session is securely invalidated.	• The current refresh token is revoked (revoked_at is set in the database). • Any subsequent request using the revoked token is rejected with 401 Unauthorised. • The access token is discarded client-side on logout.
US1.7	As a registered user, I want to reset my password via email so that I can regain access to my account if I forget my credentials.	• User submits their registered email address to initiate a reset. • A time-limited, single-use password reset link is sent to the provided address. • Following the link allows the user to set a new password without knowing the current one. • The reset token is invalidated after use or after expiry.

US1.8	As an Admin, I want to view an audit log of administrator-level actions so that I can monitor system access and detect unauthorised activity.	• Audit log displays each entry's actor, action performed, target record type and ID, and timestamp. • Log entries are read-only and cannot be modified or deleted.
US1.9	As an Admin, I want to reject and permanently delete a pending user application so that unauthorised or unwanted registrations are removed from the system.	• Admin can permanently delete any user account that is in an inactive (pending) state. • Active accounts cannot be deleted; the Admin must first deactivate the account before deletion is permitted. • Deletion permanently removes the user record and all associated tokens (refresh tokens, password reset tokens) - no soft-delete. • Admin accounts cannot be deleted via this action.

		• The deletion is recorded in the audit log (actor, action, target user ID, timestamp) before the record is removed. • * Returns 204 No Content on success.
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Data Ingestion & Geospatial Management

ID	User Story	Acceptance Criteria
US2.1	As an Analyst, I want to upload CSV files of historical data so that the risk engine has data to process.	• Supports types: <code>incidents</code>, <code>patrol_tracks</code>, <code>sightings</code>, and <code>tipoffs</code>. • File is stored and a background processing job is queued. • Validated data is persisted to PostgreSQL with PostGIS spatial indexing.
US2.2	As an Analyst, I want to use a data upload wizard with previews and validation errors so that I can fix data issues before committing.	• Wizard invokes the <code>/v1/ingestion/validate</code> endpoint to display a data preview and report errors before the user proceeds to final upload. • Displays friendly, human-readable validation errors per row.

		• Final upload is only triggered via the upload wizard once the user confirms the data is correct.
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Risk Engine & Visualization

ID	User Story	Acceptance Criteria
US3.1	As a Ranger or Analyst, I want to view a heatmap of high-risk areas so that I can identify poaching hotspots.	• Heatmap layer indicates high-risk areas based on AI Risk Engine scores. • Provides an interactive time range slider to observe changes across intervals. • Heatmap updates within 1 second of a new slider position. • Includes interactive map controls, legends, and data filters.
US3.2	As a Ranger or Analyst, I want to see the reasoning behind a risk score so that I can understand why an area is marked high-risk.	• Explainability panel communicates the specific feature contributions for a selected cell. • Clearly presents confidence levels without overclaiming model certainty.

US 3.3	As a Ranger or Analyst, I want the map to display only data my role is authorised to access so that I am not exposed to data outside my permissions.	• All map data is served via authenticated API endpoints with server-side RBAC enforcement. • Access restrictions are applied per endpoint regardless of client state. • No unauthorised data is returned in any API response.
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Field Operations & Offline PWA

ID	User Story	Acceptance Criteria
US4.1	As a Ranger, I want to submit field reports offline via the PWA so that I can capture data in areas without connectivity.	• PWA interface supports offline data capture for incidents and sightings. • All reports capture: report type (incident or sighting), location (lat/lon), occurred_at, and description. • Incident reports additionally capture incident type and severity level (low / medium / high). • Sighting reports additionally capture species name and an optional animal count. • Each submitted field report creates a corresponding geospatial event of the matching type. • Reports are stored locally in IndexedDB when offline.

		• Reports sync automatically to the server within 60 seconds of connectivity being restored. • Supports optional photo uploads to MinIO via pre-signed URL.
US4.2	As a Ranger, I want to edit and delete my field reports so that I can correct mistakes and manage my submissions.	• Ranger can edit the common fields of their own reports: description, location, occurred_at, and photos. • For incident reports, the ranger can additionally edit incident type and severity level. • For sighting reports, the ranger can additionally edit species name and animal count. • Edits made offline are stored locally and synced on reconnection.

		• Ranger can delete their own reports; deleted reports are soft-deleted and no longer visible in the UI. • A Ranger cannot edit or delete another Ranger's reports. • Changes are reflected in the UI immediately after submission.
US4.3	As a Ranger, I want to generate optimal patrol routes based on risk and resource constraints so that I can patrol safely and efficiently.	• Planner accepts starting point, maximum patrol time, and fuel constraint as inputs. • System consumes the Risk Heatmap to propose routes prioritising high-risk coverage. • System generates multiple alternative routes per planning request, grouped by a shared request ID.

		• Provides a side-by-side route comparison interface showing risk coverage, estimated time, and estimated fuel consumption for each alternative. • Route suggestions are returned within 5 seconds.
US4.4	As a Ranger or Analyst, I want to view my previously generated patrol routes so that I can review past plans without regenerating them.	• Authenticated Rangers and Analysts can retrieve a list of their own previously generated patrol route requests, grouped by request ID. • Each route record displays its start point, resource constraints, suggested path, risk coverage, estimated time, and estimated fuel consumption.
US4.5	As a Ranger, I want my offline field reports to sync without silent data loss so that I can trust that what I captured in the field is accurately reflected in the system.	• When connectivity is restored, all locally queued reports are automatically submitted to the server.

		• If a report was edited both offline and on the server, the conflict is resolved deterministically - no record is silently overwritten. • Soft-deleted records made offline are correctly propagated; they do not reappear after sync. • The Ranger is not required to take any manual action to trigger or resolve a sync.
US4.6	As an Admin, I want to edit or delete any field report so that I can correct data quality issues and manage submissions across all rangers.	• Admin can edit the common fields (description, location, occurred_at, photos) of any field report regardless of who submitted it. • For incident reports, the Admin can additionally edit incident type and severity level.

		• For sighting reports, the Admin can additionally edit species name and animal count. • Admin can soft-delete any field report; deleted reports are retained for audit purposes but are no longer visible in the UI.
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Community Tip-off

ID	User Story	Acceptance Criteria
US5.1	As a Community Liaison, I want to submit a tip-off about a suspected incident or sighting so that rangers can be informed of potential threats.	• Form captures report type (incident or sighting), location, occurred_at, description, and optional photos. • Incident tip-offs additionally capture incident type and severity level. • Sighting tip-offs additionally capture species name and optional animal count. • occurred_at must not be in the future. • Submission is associated with the authenticated Community Liaison account. • Returns 201 Created on success.

Dashboards & Analytics

ID	User Story	Acceptance Criteria
US6.1	As an Analyst or Admin, I want to view operational dashboard cards so that I can monitor patrol coverage, field report trends, and system activity at a glance.	• Dashboard displays cards for patrol coverage, field report counts, and recent trends. • Data is up to date with the latest synced records.
US6.2	As an Analyst, I want to view basic model performance metrics so that I can assess the reliability of the AI Risk Engine.	• Dashboard displays model performance metrics such as accuracy indicators. • Metrics are clearly labelled and do not overclaim model certainty.

Use Cases

This is a summarised version of the Use Cases document. For more information, please review the Use Cases folder in the Github, which includes additional details such as when the use case starts and ends

Authentication Subsystem

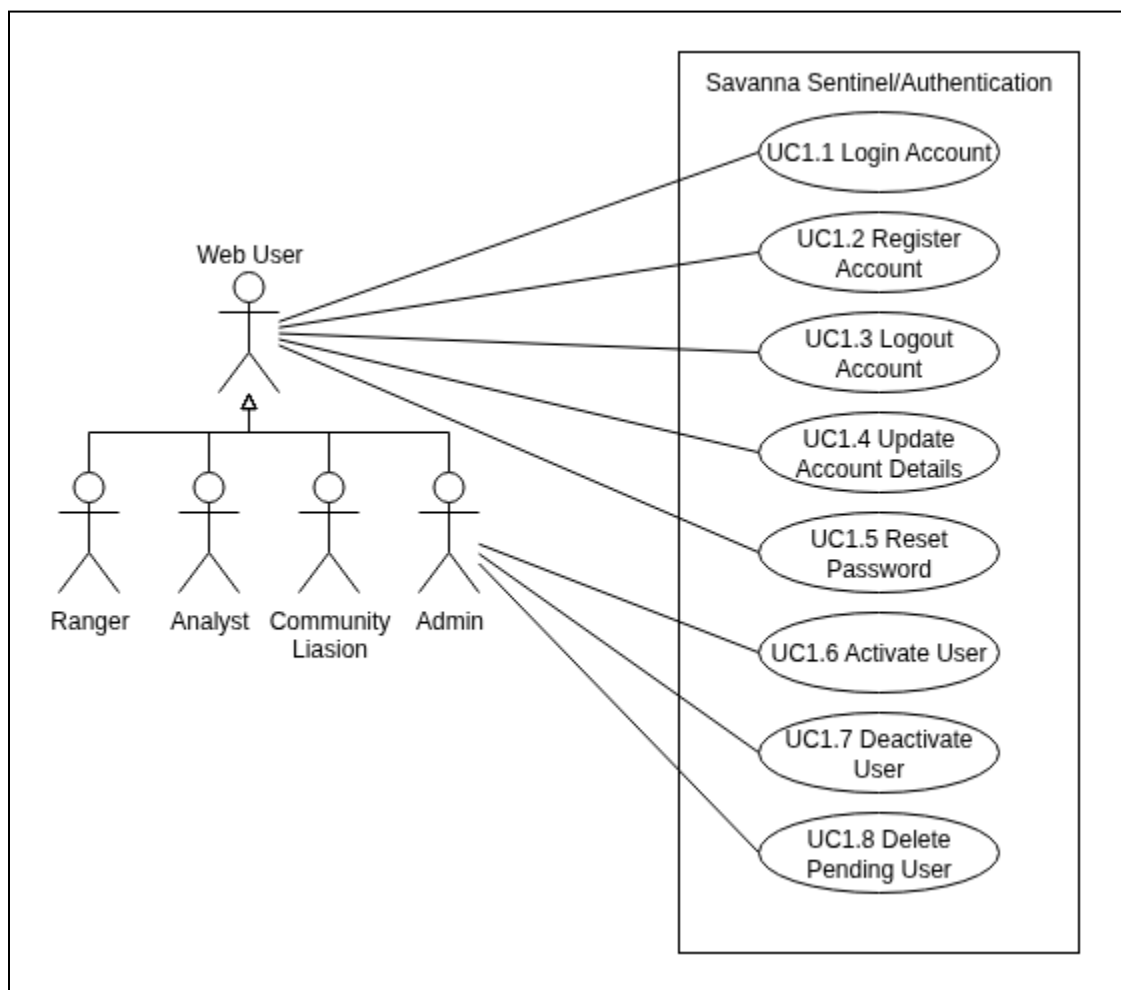


Figure 1 - Authentication Subsystem

Ingestion Subsystem

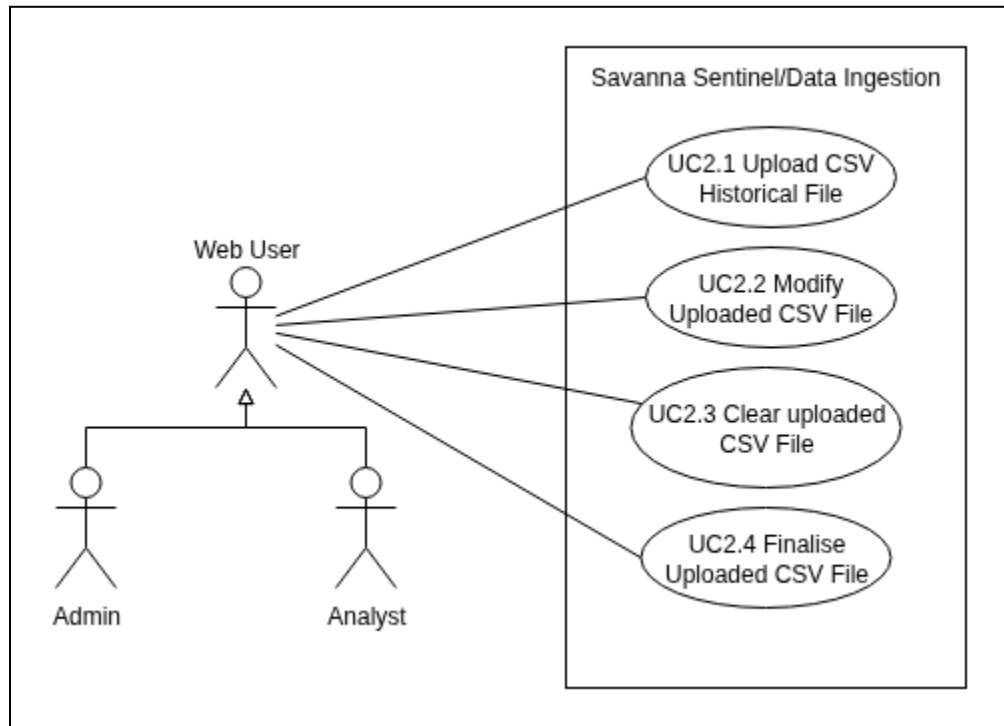


Figure 2 - Ingestion Subsystem

Risk Subsystem

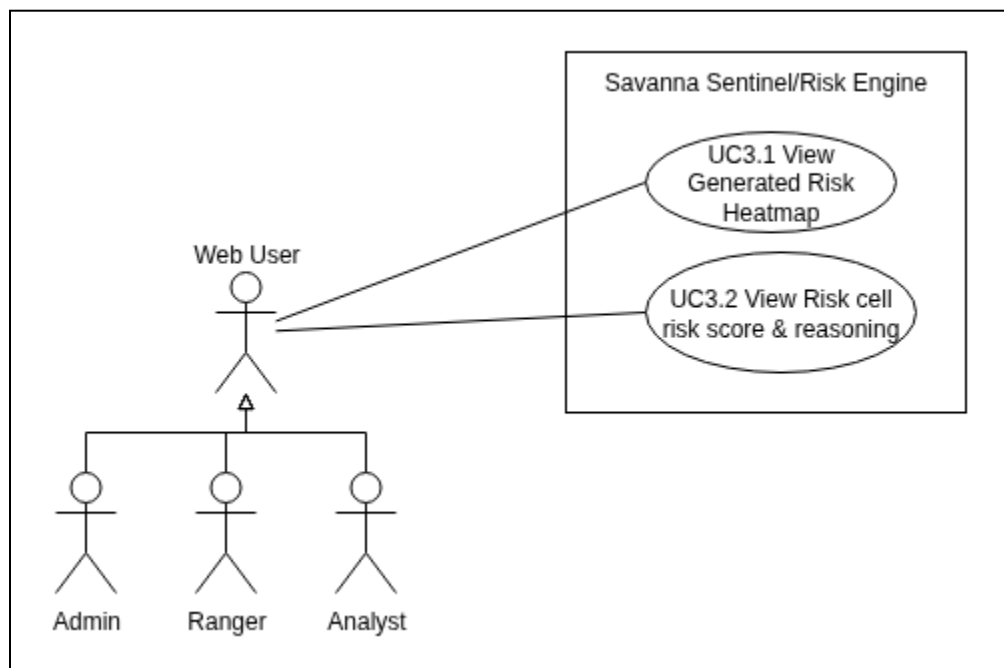


Figure 3 - Risk Subsystem

Field Report and Routes Subsystem

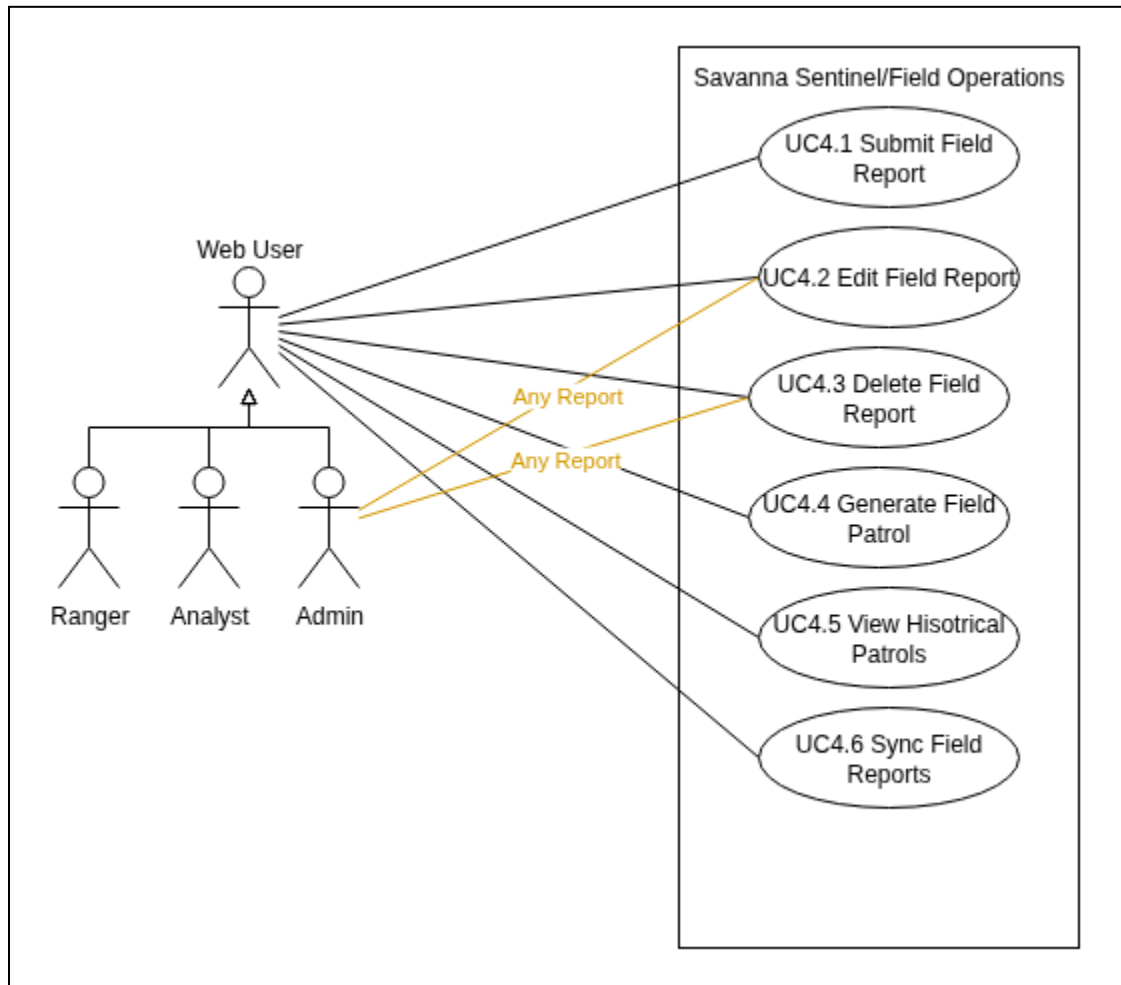


Figure 4 - Field Report and Routes Subsystem

Tip-off Subsystem

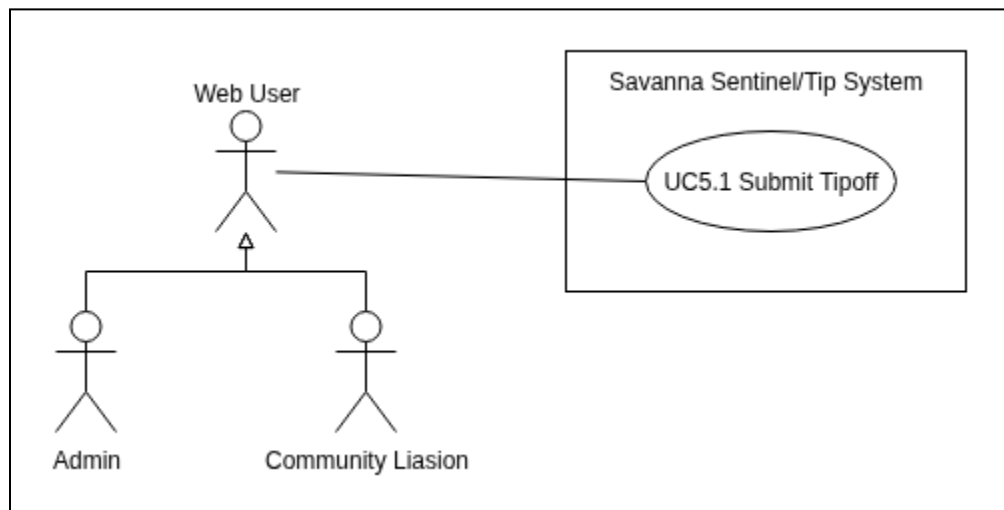


Figure 5 - Tip-off Subsystem

Dashboard Subsystem

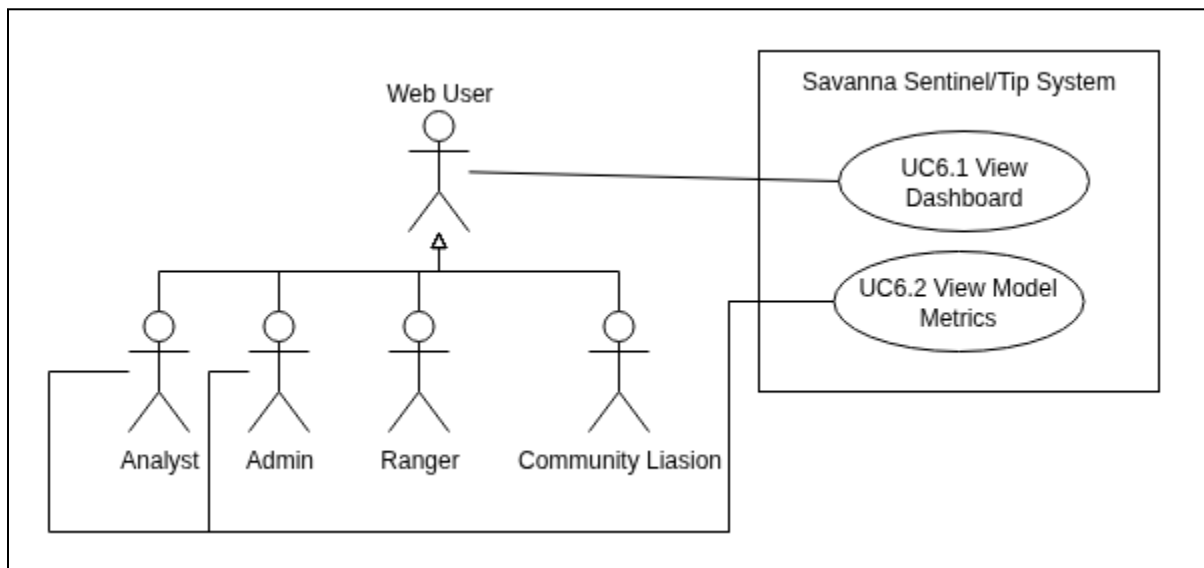


Figure 6 - Dashboard Subsystem

Admin Subsystem

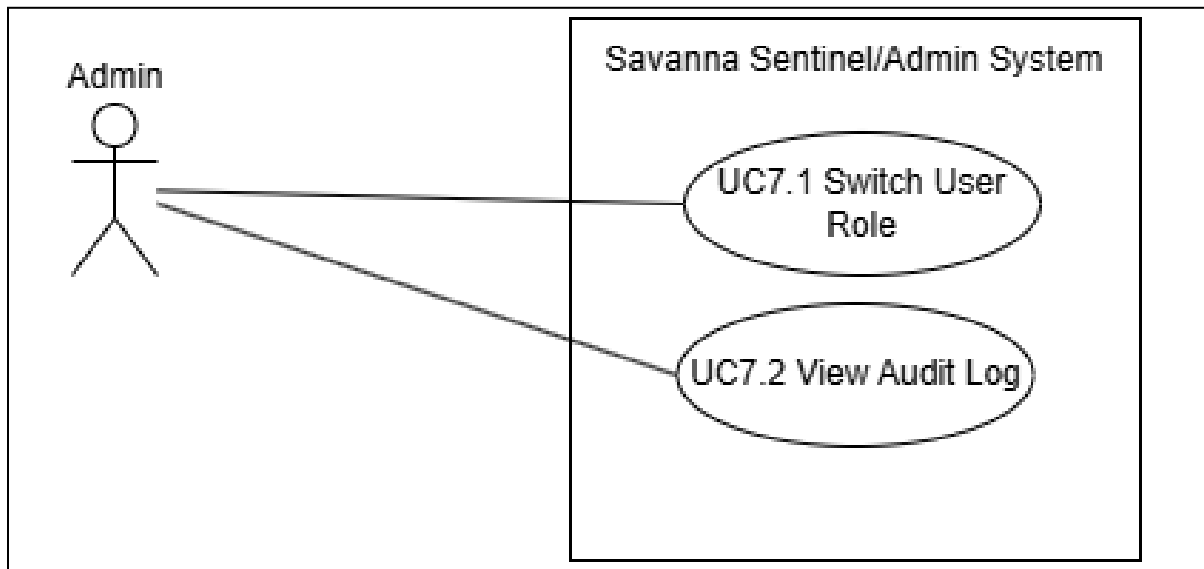


Figure 6 - Admin Subsystem

Functional Requirements

Dedicated Functional Requirements document can be found in the github docs folder

R1: User Access and Security Management

Subsystem: Auth Identity System

R1.1 Role Based Access Control (RBAC)

- **R1.1.1:** The system will support role-based access control with predefined roles including Ranger, Analyst, Community Liaison and Admin.
- **R1.1.2:** The system will enforce server-side authentication using stateless sessions (JWT).
- **R1.1.3:** The system will restrict access to specific views, features, and map layers based on the authenticated user's assigned role.

R1.2 Audit and Security

- **R1.2.1:** The system will maintain an audit log of all administrator-level actions, recording the actor's user ID, action performed, target record type and ID, and a timestamp
- **R1.2.2:** The audit log shall be readable only by Admins.
- **R1.2.3:** The audit log shall be immutable, no user, including Admins, shall be able to edit or delete an existing audit log entry.
- **R1.2.4:** The system will ensure all data in transit is encrypted via HTTPS.

R2: Data Ingestion and Validation

Subsystem: Data Ingestion Subsystem

R2.1: File Upload and Parsing

- **R2.1.1:** The system will allow Analysts to upload CSV files containing historical data.
- **R2.1.2:** The system will provide a data upload wizard that displays a preview of the uploaded data prior to commit.
- **R2.1.3:** The data upload wizard will display human-readable validation errors for any row that fails validation, without committing any data until the Analyst confirms the import.
- **R2.1.4:** The system will parse and translate uploaded data into core entities including: Incidents, Patrol Tracks, Sightings, and Tip-offs.

R2.2: Geospatial Storage

- **R2.2.1:** The system will persist validated geospatial data with spatial indexing to support efficient location-based querying.

R3: Geospatial Risk Engine and Visualization

Subsystem: Geospatial Risk Engine Subsystem

R3.1: Risk Engine Processing

- **R3.1.1:** The AI Risk Engine will process historical data via asynchronous background jobs to compute risk scores, such that the requesting user is not blocked while scores are calculated.
- **R3.1.2:** The Risk Engine will calculate risk hotspots gridded into geographic cells.
- **R3.1.3:** The Risk Engine will compute a distinct risk score per geographic cell for each defined time interval.

R3.2: Interactive Map

- **R3.2.1:** The system will display a heatmap layer on the map indicating high-risk areas based on the calculated risk scores.
- **R3.2.2:** The system will provide an interactive time range slider allowing users to select a time interval and view its corresponding risk hotspots
- **R3.2.3:** The system will provide interactive map controls including a legend describing the heatmap scale.
- **R3.2.4:** The system will provide interactive data filters allowing users to filter map layers by record type (e.g Incidents, Sightings, Tip-offs)
- **R3.2.5:** The system will enforce role-based access control on all map data API endpoints, such that each role receives only the data it is authorised to access as defined by the RBAC policy.

R4: Risk Engine Explainability

Subsystem: Risk Explainability Subsystem

R4.1: Explainability Metrics

- **R4.1.1:** The Risk Engine will generate explainability metrics using SHAP for each computed risk score.
- **R4.1.2:** The generated explainability metrics will detail the contribution of each input feature to the resulting risk score.

R4.2: Explainability Panel

- **R4.2.1:** The system will provide an explainability UI panel that communicates the specific reasons a geographic cell is designated as high risk.
- **R4.2.2:** The explainability UI will display a confidence level for each risk score.
- **R4.2.3:** The explainability UI's confidence level display must not overclaim model certainty (e.g. must not present a score as a guarantee of an event occurring)

R5: Patrol Route Planning

Subsystem: Patrol Planning Subsystem

R5.1: Constraint-Based Route Generation

- **R5.1.1:** The Patrol Planner will accept custom user parameters, including starting point, maximum time allowed and fuel usage constraints.
- **R5.1.2:** The Patrol Planner will consume the current Risk Heatmap and the supplied resource constraints to generate multiple alternative patrol routes per planning request.
- **R5.1.3:** All route alternatives generated from a single planning request will share a common request ID and will be returned together to the requesting user.

R5.2: Route Retrieval

- **R5.2.1:** The system will allow authenticated Rangers and Analysts to retrieve their previously generated patrol route requests.
- **R5.2.2:** Retrieved route alternatives will be grouped by their shared request ID.

R5.3: Route Evaluation

- **R5.3.1:** The system will provide a side-by-side route comparison interface allowing rangers to evaluate proposed routes.
- **R5.3.2:** The route comparison interface will display, for each route option, the estimated risk coverage, estimated time and estimated fuel consumption.

R6: Field Reporting and Offline Sync

Subsystem: Field Reporting Subsystem

R6.1: Mobile Offline Capture

- **R6.1.1:** The system will provide a Progressive Web App (PWA) interface for rangers to capture Field Reports.
- **R6.1.2:** The system will allow Rangers to input field report data locally on the device when operating without network connectivity.
- **R6.1.3:** A Ranger may submit multiple field reports during a single patrol.
- **R6.1.4:** Each field report shall be classified as either an incident report or a sighting report.
- **R6.1.5:** An incident report shall capture an incident type and a severity level, in addition to the common fields.
- **R6.1.6:** A sighting report shall capture a species name and an optional animal count, in addition to the common fields.
- **R6.1.7:** Each submitted field report will create a corresponding geospatial event of the matching type (Incident or Sighting).
- **R6.1.8:** Offline field report data shall be persisted to the device using IndexedDB until synchronisation occurs.
- **R6.1.9:** The system will optionally support capturing photo evidence associated with a field report.
- **R6.1.10:** The system will upload captured photo evidence to S3-compatible storage (MinIO) via pre-signed URLs.

R6.2: Field Report Management

- **R6.2.1:** The system will allow Rangers to edit their own previously submitted field reports, both online and offline.
- **R6.2.2:** For all field reports, the editable common fields are description, location, occurred_at and associated photos.
- **R6.2.3:** For incident reports specifically, incident type and severity shall additionally be editable.
- **R6.2.4:** For sighting reports specifically, species and count shall additionally be editable.
- **R6.2.5:** The system will allow Rangers to delete their own previously submitted field reports.
- **R6.2.6:** Deleted field reports will be soft-deleted and retained in the database for audit purposes.
- **R6.2.7:** The system will allow Admins to edit any field report regardless of the submitting Ranger, with the same field-level permissions defined in R6.2.2–R6.2.4.
- **R6.2.8:** The system will allow Admins to delete any field report regardless of the submitting Ranger.

R6.3: Synchronisation and Conflict Handling

- **R6.3.1:** The system will allow Admins to delete any field report regardless of the submitting Ranger.
- **R6.3.2:** The Sync Service will detect database conflicts arising from offline data uploads and offline deletions.
- **R6.3.3:** The Sync Service will resolve detected conflicts deterministically, such that no record is silently overwritten.
- **R6.3.4:** Soft-deleted records created offline shall be correctly propagated to the central database upon sync.

R7: Community Tip-off Submission

Subsystem: Community Tip-off Subsystem

R7.1: Tip-off Capture

- **R7.1.1:** The system will allow Community Liaisons to submit tip-offs describing an observed incident or sighting.
- **R7.1.2:** Every tip-off shall capture report type, location, occurred_at and description.
- **R7.1.3:** The system will optionally allow photo evidence to be attached to a tip-off.
- **R7.1.4:** An incident tip-off shall additionally capture an incident type and a severity level.
- **R7.1.5:** A sighting tip-off shall additionally capture a species name and an optional animal count.
- **R7.1.6:** The system will reject a tip-off submission if the occurred_at field is set to a future date or time.
- **R7.1.7:** The system will associate each submitted tip-off with the submitting Community Liaison's user account.

R8: Dashboards and System Analytics

Subsystem: Analytics & Dashboard Subsystem

R8.1: Operational Dashboards

- **R8.1.1:** The system will display dashboard cards summarising patrol coverage
- **R8.1.2:** The system will display dashboard cards summarising Field Report findings.
- **R8.1.3:** The system will display dashboard cards summarising operational trends over time.
- **R8.1.4:** The system will display basic model performance metrics to Analysts.
- **R8.1.5:** Displayed model performance metrics shall be clearly labelled and must not overclaim model certainty.

R9: Account and Identity Management

Subsystem: Account & Identity Management Subsystem

R9.1: Authentication Flow

- **R9.1.1:** The system will allow only activated user accounts to log in.
- **R9.1.2:** The system will deny login access to accounts pending Admin activation.
- **R9.1.3:** The system will return a deliberately vague error response for any failed login attempt, such that the response does not reveal whether the email address exists, the password is incorrect or the account is inactive.
- **R9.1.4:** The system will allow users to log out, securely invalidating their current active session.
- **R9.1.5:** The system will allow clients to obtain a new access token using a valid, non-revoked refresh token without requiring the user to log in again.
- **R9.1.6:** The system will return a 401 Unauthorised response if a client attempts to use an invalid or revoked refresh token.
- **R9.1.7:** The system will provide a password recovery mechanism whereby a user submits their registered email address and the system sends a time-limited, single-use password reset link to that address.
- **R9.1.8:** Following a valid password reset link will allow the user to set a new password without providing their current one.
- **R9.1.9:** The system will invalidate a password reset token after its first use or after its expiry, whichever occurs first.
- **R9.1.10:** The system will request a pin to be entered that will be sent to the user's email when the account is identified to have admin role permissions.

R9.2: User Registration

- **R9.2.1:** The system will allow prospective users to register for an account by providing a username, email address, password, first name and last name.
- **R9.2.2:** At registration, the system will require the prospective user to select their intended role from the available options: Ranger, Analyst, or Community Liaison.
- **R9.2.3:** All newly registered accounts will be created in an inactive state.
- **R9.2.4:** The system will require explicit activation by an Admin before a newly registered user can log in.

R9.3: User Account Administration

- **R9.3.1:** The system will allow an Admin to view a comprehensive list of all system users and their current statuses, including pending activation requests.
- **R9.3.2:** The system will allow an Admin to activate a user account.
- **R9.3.3:** The system will allow an Admin to deactivate a user account.
- **R9.3.4:** The system will allow an Admin to modify the role of any non-Admin user account.
- **R9.3.5:** The system will prevent any Admin from activating, deactivating, or modifying the role of an account that holds the Admin role.
- **R9.3.6:** The system will allow an Admin to permanently delete a user account that is in an inactive (pending) state.
- **R9.3.7:** The system will prevent the deletion of any user account that is in an active state, an Admin must first deactivate an account before deletion is permitted.
- **R9.3.8:** Upon deletion, the system will permanently remove the user record and all associated tokens (refresh tokens, password reset tokens) from the database, with no soft-delete applied.
- **R9.3.9:** The system will prevent any Admin from deleting an account that holds the Admin role.
- **R9.3.10:** The system will record a user account deletion in the audit log, including actor ID, action, target user ID, and timestamp, before the record is removed.

R9.4: Profile Management

- **R9.4.1:** The system will allow users to view their own basic profile information.
- **R9.4.2:** The system will allow users to update their own basic profile information.
- **R9.4.3:** The system will allow users to securely update their own password.
- **R9.4.4:** A password change request shall require the user's current password for verification before the change is accepted.
- **R9.4.5:** On a successful password change, the system will revoke all existing refresh tokens associated with that account.

R10: General Application Features

Subsystem: Core Application Services Subsystem

R10.1: Data Entry and Validation

- **R10.1.1:** The system will enforce client-side form validation on all user inputs.
- **R10.1.2:** The system will enforce server-side form validation on all user inputs, independent of client-side validation, to prevent malformed data
- **R10.1.3:** The system will require user passwords to be a minimum of 8 characters in length.
- **R10.1.4:** The system will display clear, human-readable error messages when form validation fails.

Non-Functional Requirements

Dedicated Non-Functional Requirements document can be found in the github docs folder

Core Non-Functional Requirements

Our system targets the following non-functional requirements

- **Security:** All sensitive user data should be encrypted using AES-256 encryption, and the system should enforce multi-factor authentication for administrative accounts.
- **Reliability:** The system should achieve 99.9% uptime and recover from critical failures within 5 minutes.
- **Performance:** The system should respond to user requests within 2 seconds for 95% of requests under normal operating conditions and support at least 60 concurrent users.
- **Availability:** The system should be available 24/7, excluding scheduled maintenance periods not exceeding 2 hours per month.
- **Maintainability:** New features or bug fixes should be deployable within 2 hours, and the codebase should maintain at least 80% automated test coverage.

Quantifications

Security

Mozilla HTTP Observatory Badge on the main README page.

Reliability

Uptime Badge on the main README page.

Performance

K6 Performance test with ~90 Concurrent users performing various tasks in estimated distributions, with a badge on the main README stating the p(95) response time

Availability

Automated restarts of crashing containers through docker, with a current website status badge on the README

Maintainability

- Backend CI Status Badge
- Frontend CI Status Badge
- CodeCov Badge

Domain Model

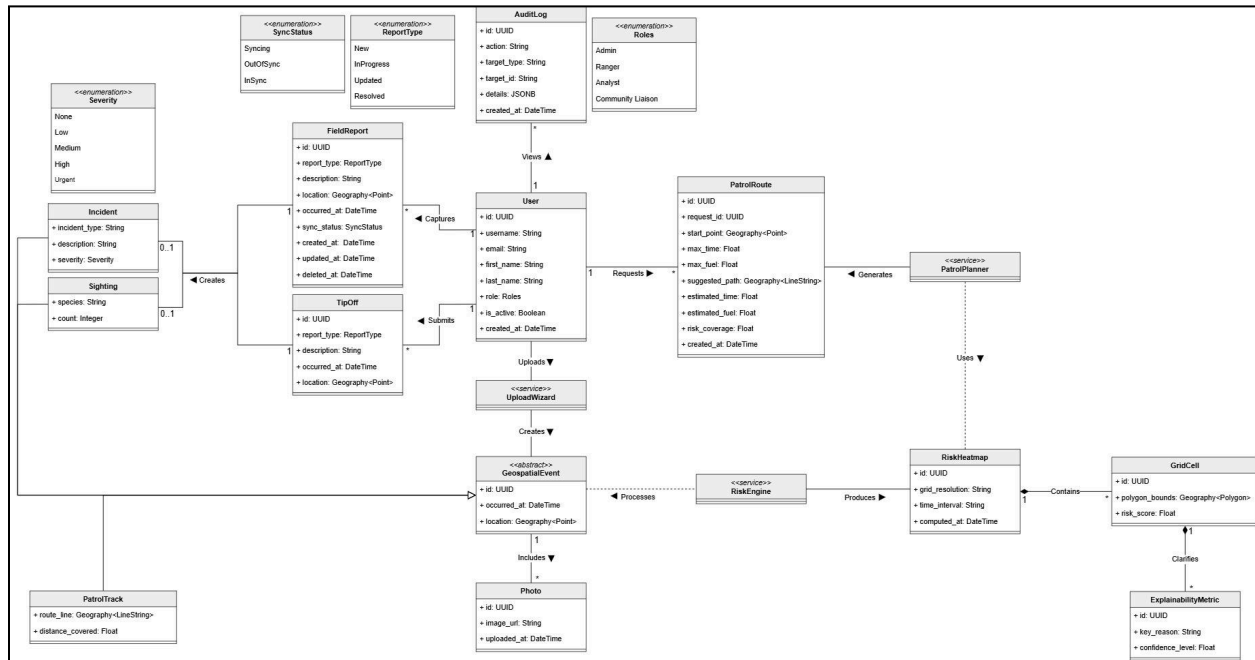


Figure 7 - Domain Model